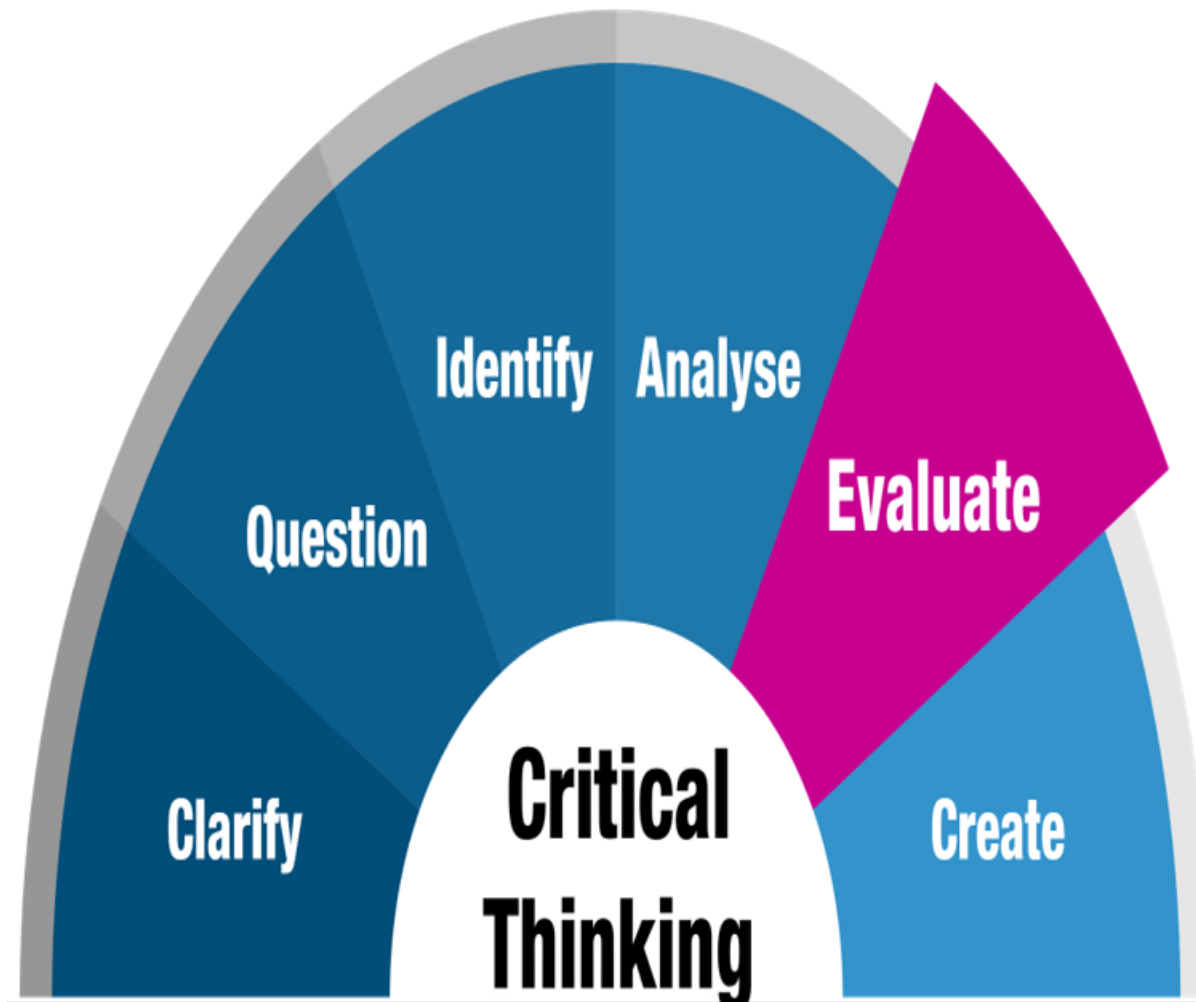




Evaluating Arguments

Evaluating **Deductive** Arguments:
Validity & Soundness

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Lesson Objectives:

- **After the successful accomplishment of this lesson, you will be able to:**
 - Understand how to evaluate deductive arguments in terms of validity and soundness.
 - Recognize the relationship between truth value and validity.
 - Understand how to evaluate inductive arguments in terms of strength and cogency.
 - Recognize the relationship between truth value and strength.



Evaluating Deductive Arguments: Validity and Soundness

- Valid deductive argument

It is an argument where the premises are assumed true, it is impossible for the conclusion to be false.

In such arguments, the conclusion follows with strict necessity from the premises.

So ,

The premises is true



The conclusion must be true



Evaluating Deductive Arguments: Validity, Truth, and Soundness

- Invalid deductive argument

It is an argument where if the premises are assumed true, it is possible for the conclusion to be false.

In these arguments, the conclusion does not follow with strict necessity from the premises,

So ,

The premises is true



The conclusion may be false



Let's practice

Practice
Makes
Perfect

Example-1:

*All men are mammals.
All bulls are men.
Therefore, all bulls are mammals.*

Type : valid argument.

Because the conclusion actually followed from the premises with a strict necessity.

If all men are assumed as mammals and bulls as men, then it is **impossible for bulls not be mammals.**



Example-2:

*All philosophers are rational.
Socrates was rational.
Therefore, Socrates was a philosopher.*

Type : invalid argument

Because the conclusion did not actually follow from the premises with a strict necessity, even though it is claimed to.

if we assume that all philosophers rational and Socrates is rational, **it is not** actually impossible for Socrates **not be** a philosopher.



please note

- The idea that any deductive argument having actually true premises and a false conclusion is invalid may be the most important point in the entire system of deductive logic.

True premises and False conclusion= invalid argument

- **Example-1 (Invalid):**
- *All biologists are scientists. (Tp)*
- *John Nash was a scientist. (Tp)*
- *Therefore, John Nash was a biologist. (Fc)*



Trick or Treats



Answer the following questions



A. 1. All dogs bark. 2. Charlie is a dog. Therefore, Charlie barks.

1. Valid
2. Invalid

B. 1. Almost all dogs can play the piano. 2. Charlie is a dog. Therefore, Charlie can play the piano.

1. Valid
2. Invalid

C. 1. All dogs bark. 2. Charlie barks. Therefore, Charlie is a dog.

1. Valid
2. Invalid



Answer

a. Valid b. Invalid c. invalid



Answer the following questions



A. 1. If there is no gas in the car, then it won't start. 2. The car started. Therefore, there is gas in the car.

1. Valid
2. Invalid



B. 1. The card you picked is either a Jack, a Queen, or a King. 2. The card you picked is not a Jack. Therefore, the card you picked is either a Queen or a King.

1. Valid
2. Invalid



Answer

a. Valid b. valid



Answer the following questions



A. 1. Successful candidates for this job must have either a Master's degree or five years of work experience. 2. Johnny has a Master's degree. 3. Johnny got the job. Therefore, Johnny does not have five years of work experience.

1. Valid
2. Invalid

Answer

a. Invalid



Answer the following questions



- **True or False:**

1. A valid argument can have false premises and a true conclusion.

- A. True.
- B. False.

2. An invalid argument can have true premises and a true conclusion.

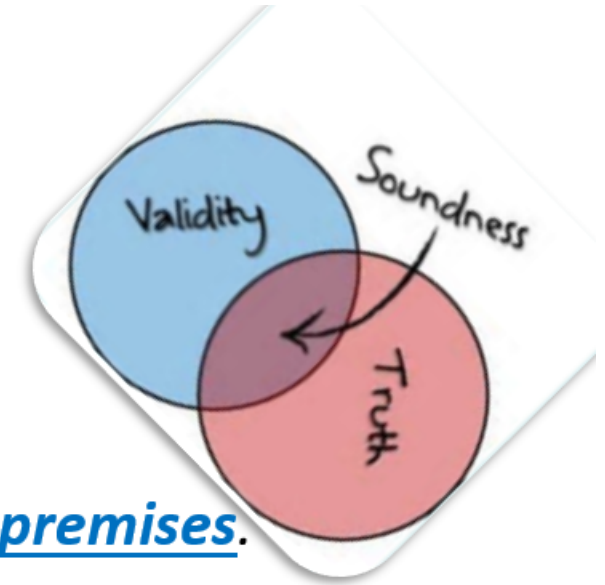
- A. True.
- B. False.



Answer

- a. true
- b. true

Deduction and Soundness



Sound argument

It is a deductive argument that is valid and has all true premises.

Unsound argument

It is a deductive argument that is either valid with one or more false premises, or invalid, or both.

Sound Argument = A valid argument + All true premises